

Crop Production and Economic Impact in the U.S.

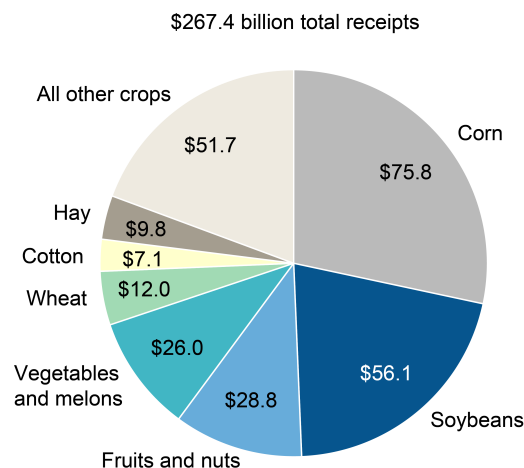
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1 Introduction

According to the Economic Research Service at the U.S. Department of Agriculture, the top five U.S. crop cash receipts in 2023 were corn, soybeans, fruits and nuts, vegetables and melons, and wheat [1].

2023 U.S. crop cash receipts



Note: Components may not sum to total because of rounding. Data as of February 6, 2025.
Source: USDA, Economic Research Service, Farm Income and Wealth Statistics.

Figure 1: Image source: USDA ERS

To further understand how these cash crops contribute to the overall economy, we are interested in the relationship between crop production and GDP. Thus, our research question is:

Is higher production of major cash crops (corn, soybeans, and wheat) significantly associated with higher county-level GDP?

Understanding how agricultural productivity connects to economic performance can help inform resource allocation in rural economies and guide agricultural policy at both the state and federal levels.

2 Data and Exploratory Data Analysis

Our datasets come from the U.S. Department of Agriculture (USDA) [2] for crop data, and the Bureau of Economic Analysis (BEA) [3] for GDP data. They cover all U.S. counties reporting 2022 agricultural data for at least one of the three major crops - corn, soybeans, or wheat - along with corresponding GDP figures. Since the most recent complete dataset on crop production is from 2022, we focus our analysis on that year.

Because we are interested in the economic implications of agricultural productivity, our outcome variable is county GDP (measured in thousands of chained 2017 U.S. dollars), as it reflects overall economic performance. Our predictors, based on available data, include:

- Production (in bushels),
- Area harvested (in acres),
- Sales (in U.S. dollars), and
- Crop type (corn, soybeans, or wheat).

During our exploratory analysis, we found a clear positive association between sales and production (Figure 2). Among the three crops, corn requires the most production to generate a given amount of sales, while soybeans require the least, reflecting their relative market prices.

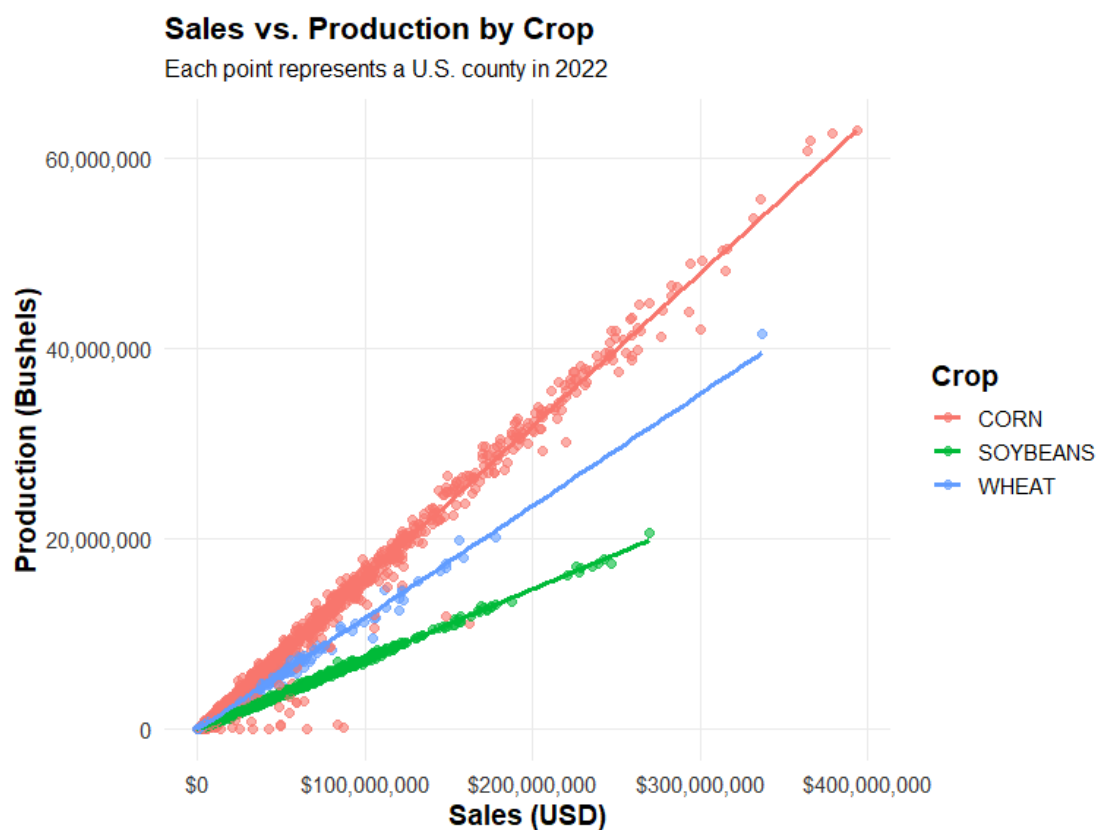


Figure 2: Sales vs. Production by Crop. Each point represents a county-crop observation. Soybeans yield higher sales per bushel compared to corn and wheat, suggesting price differences.

We also observed that higher harvested area is associated with higher production across all crops (Figure 3). Among them, corn appears the most area-efficient, while wheat requires the most acreage to produce the same quantity of output.

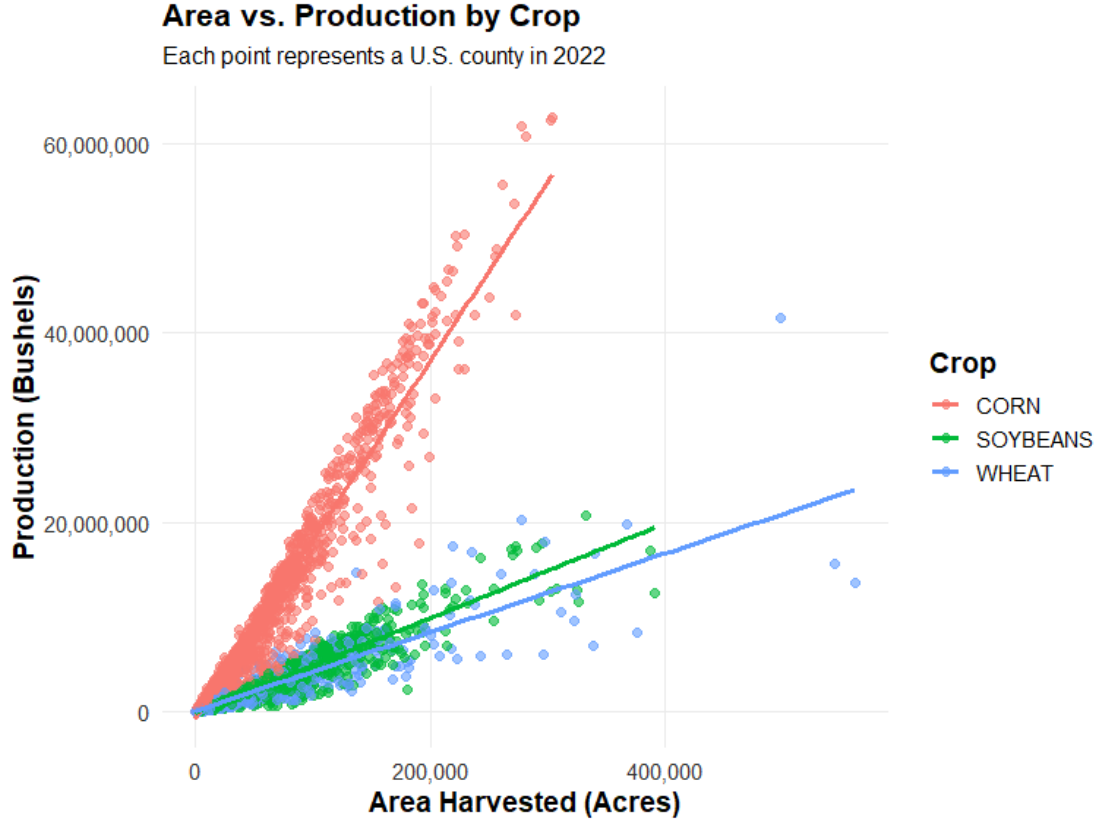


Figure 3: Area Harvested vs. Production by Crop. Corn demonstrates the highest production efficiency per acre, while wheat shows lower efficiency.

3 Methods

As shown in the previous section, there is a clear association between production, area harvested, and sales. Therefore, we include all three as predictors in our multiple linear regression model:

$$E[\text{GDP} \mid \text{production, area, sales, crop}] = \beta_0 + \beta_1 \cdot \text{production} + \beta_2 \cdot \text{area} + \beta_3 \cdot \text{sales} + \beta_4 \cdot \text{crop}$$

We also include crop type as a categorical variable to adjust for structural differences in economic impact across crops, since different crops vary in both market value and yield characteristics.

Because we are primarily interested in the effect of crop production on county GDP, we frame the following hypothesis test:

- Null hypothesis (H_0): There is no association between production and GDP.

$$H_0 : \beta_1 = 0$$

- Alternative hypothesis (H_A): There is an association between production and GDP.

$$H_A : \beta_1 \neq 0$$

This model allows us to assess whether production contributes significantly to explaining variation in GDP, while controlling for harvested area, sales, and crop type.

4 Results

After fitting the model, we obtain the following regression coefficients:

Predictor	Estimate	Std_Error	t_value	p_value
(Intercept)	6,313,000	397,000	15.90	< 0.001
Production	-0.236	0.131	-1.80	0.0726
Area	-43.37	9.527	-4.55	< 0.001
Sales	0.049	0.024	2.02	0.0437
Crop: Soybeans	-869,000	600,200	-1.45	0.1477
Crop: Wheat	-335,500	561,800	-0.60	0.5504

Figure 4: Regression coefficients for the model: $GDP = \text{Production} + \text{Area} + \text{Sales} + \text{Crop}$.

These results suggest that, holding area, sales, and crop type constant, an increase of 1 bushel in crop production is associated with an average decrease of \$236 in GDP. However, this effect is not statistically significant ($p = 0.0726$).

In contrast, area harvested is statistically significant ($p < 0.001$), indicating a negative association between area and GDP after adjusting for other variables. Specifically, for each additional acre harvested, county GDP decreases by an average of \$43,370.

Based on the p-values, we fail to reject the null hypothesis regarding the effect of production on GDP. However, we find sufficient evidence to support a negative relationship between harvested area and GDP. It is plausible that an increase of one acre in harvested area is associated with a decrease in GDP between \$24,316 and \$62,424.

5 Conclusion

This study examined whether higher production of major cash crops - corn, soybeans, and wheat - is associated with higher county-level GDP. While we found no significant relationship between production and GDP, we did find a significant negative association between harvested area and GDP. On average, increasing harvested area by one acre is linked to a GDP decrease of \$24,316 to \$62,424, after adjusting for other variables.

This may reflect the fact that agriculture-intensive counties are fewer and often more rural, while non-agricultural land use may contribute more to local GDP. It might be economically wise for

some counties to focus less on agriculture, while a smaller number of high-output counties continue producing enough food for the nation.

Still, agriculture isn't just about GDP. Food is essential to life, and these crops support broader systems like livestock production. So while GDP is a useful metric, decisions around agriculture must also consider food security and sustainability.

This analysis is based on 2022 data and cannot establish causality. Future studies could include time trends or more detailed measures like yield or regional effects.

References

- [1] U.S. Department of Agriculture Economic Research Service. *Top U.S. Crop Cash Receipts, 2023*. Accessed: 2025-04-23. 2025. URL: <https://www.ers.usda.gov/data-products/chart-gallery/chart-detail?chartId=76946>.
- [2] U.S. Department of Agriculture National Agricultural Statistics Service. *Quick Stats Database*. Accessed: 2025-04-23. 2025. URL: <https://quickstats.nass.usda.gov/>.
- [3] U.S. Bureau of Economic Analysis. *Gross Domestic Product by County, Metro, and Other Areas*. Accessed: 2025-04-23. 2025. URL: <https://www.bea.gov/data/gdp/gdp-county-metro-and-other-areas>.